27 Ans: B

The highest energy level the hydrogen atom can reach is n = 3 if max KE is 12.5 eV.
The following transitions and the corresponding wavelengths are:
$\lambda = hc/E = [(6.63 \times 10^{-34})(3.00 \times 10^{8})/(1.60 \times 10^{-19})]/\Delta E$ in eV

n = 3 to n = 2: $\lambda = 1.24 \times 10^{-6} /(3.40 - 1.53) = 6.65 \times 10^{-7}$ m (red)
n = 3 to n = 1: $\lambda = 1.24 \times 10^{-6} /(13.6 - 1.53) = 1.03 \times 10^{-7}$ m (not visible, outside 400nm to 700nm)
n = 2 to n = 1: $\lambda = 1.24 \times 10^{-6} /(13.6 - 3.40) = 1.22 \times 10^{-7}$ m (not visible, outside 400nm to 700nm)

28 Ans: A